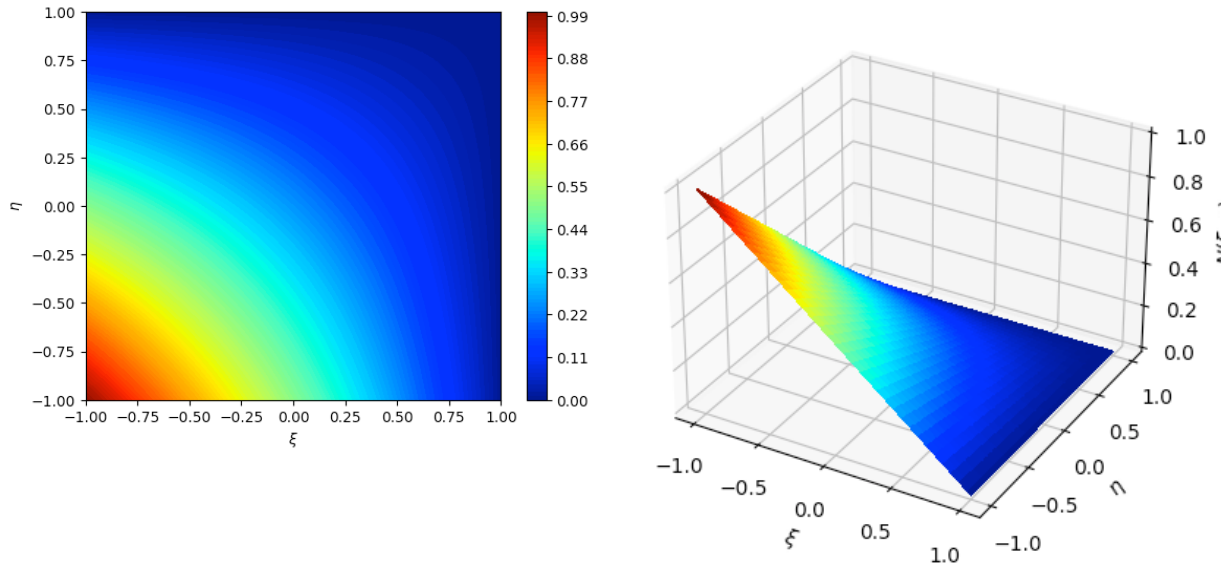
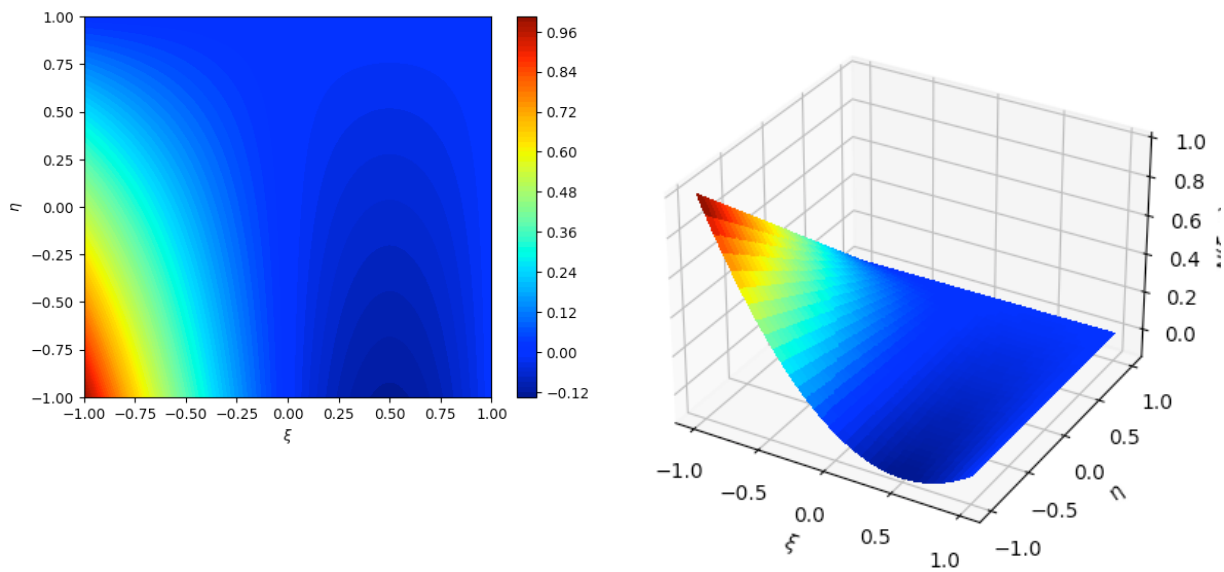
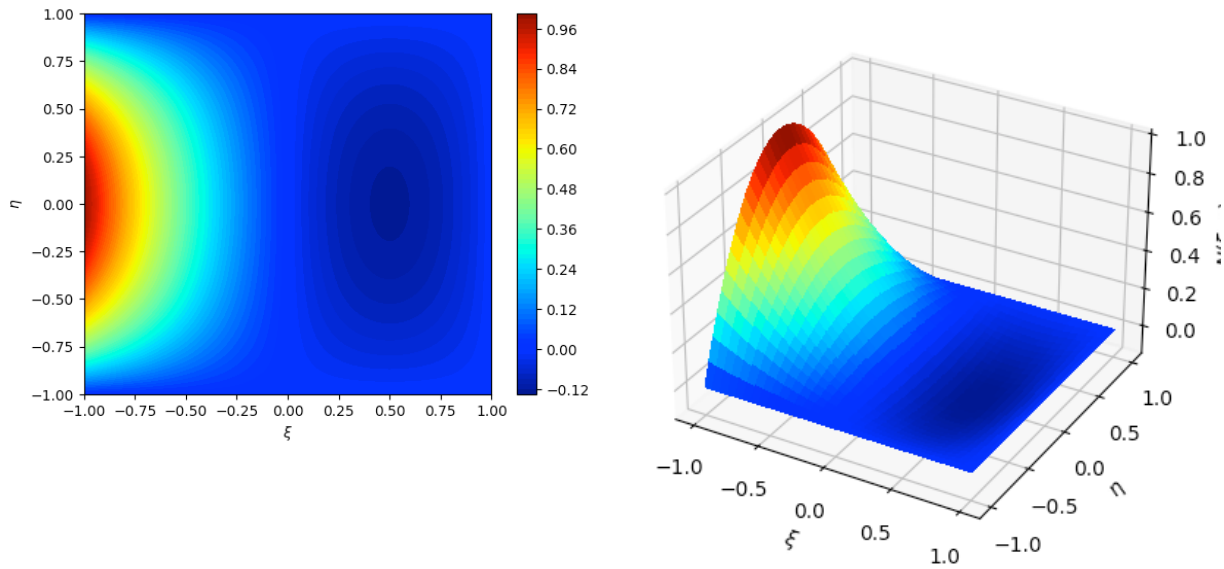
Problem 1



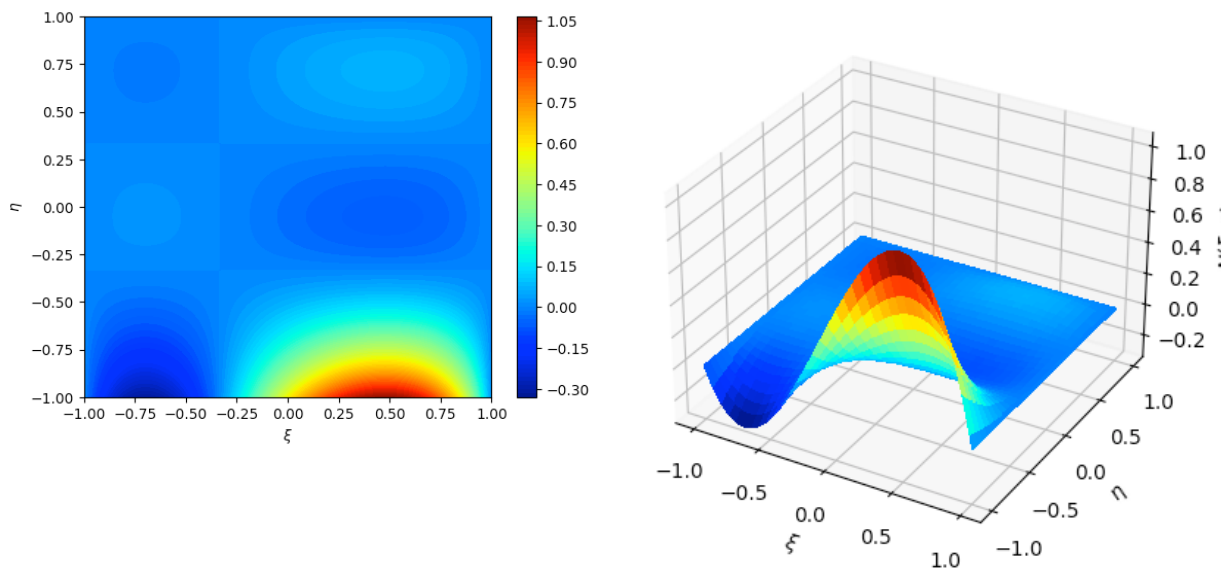
Two dimensional basis functions of the polynomial degree [1,1]



Two dimensional basis functions of the polynomial degree [2,1]

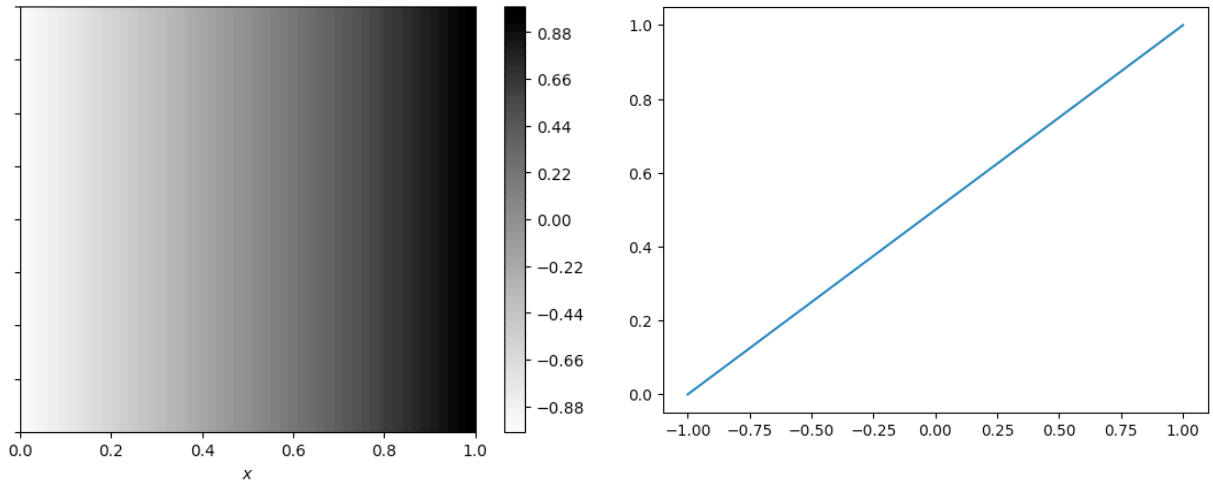


Two dimensional basis functions of the polynomial degree [2,2]

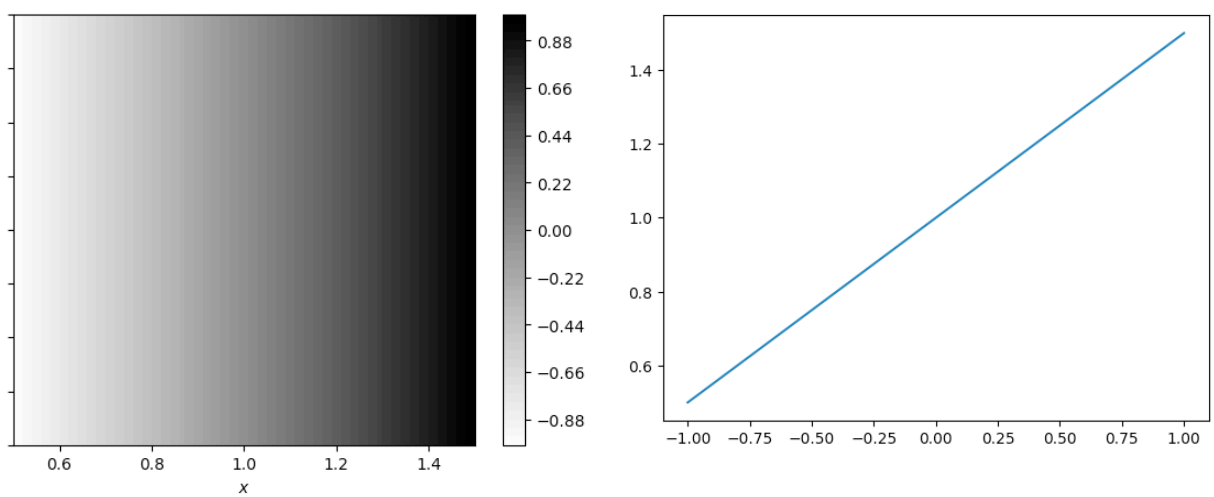


Two dimensional basis functions of the polynomial degree [3,3]

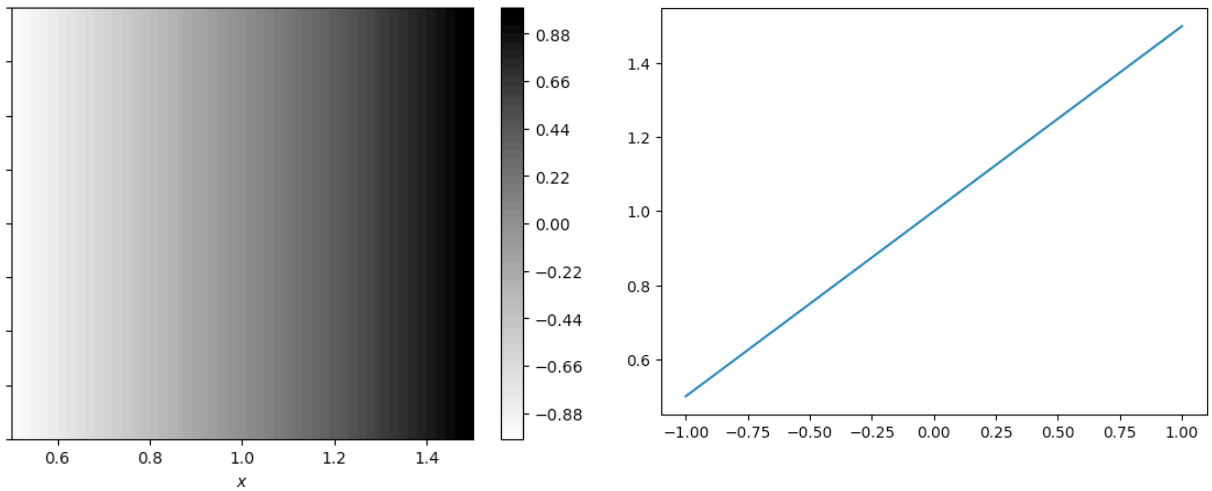
Problem 2



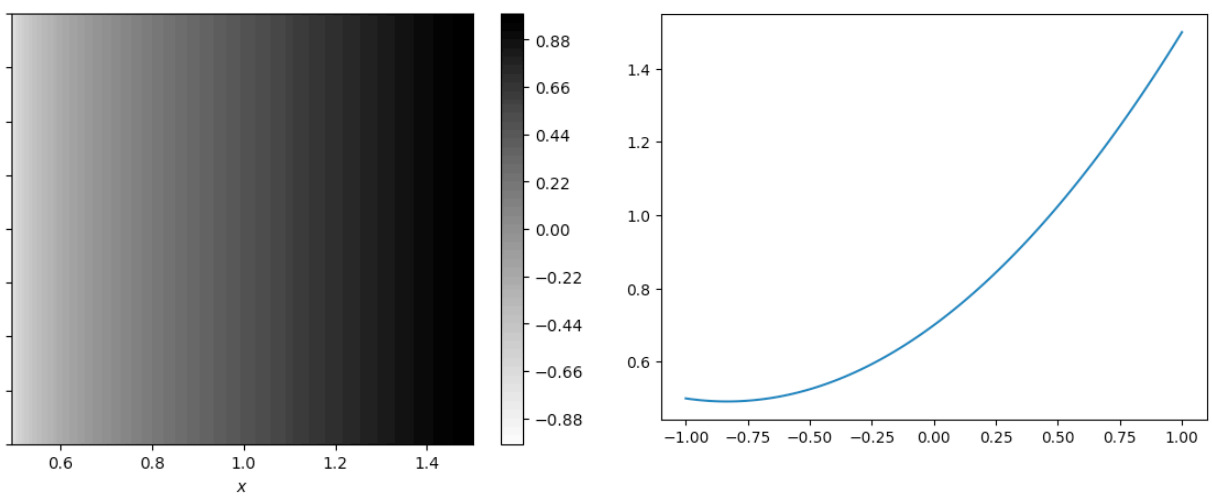
Mapping from the parent domain into a spacial domain for the input:
$p = 1$, $\{xa\} = \{0, 1\}$, $\{\xi a\} = \{-1, 1\}$



Mapping from the parent domain into a spacial domain for the input:
$p = 2$, $\{xa\} = \{0.5, 1, 1.5\}$, $\{\xi a\} = \{-1, 0, 1\}$



Mapping from the parent domain into a spacial domain for the input:
$p = 2, \{xa\} = \{0.5, 0.7, 1.5\}, \{\xi a\} = \{-1, -0.6, 1\}$



Mapping from the parent domain into a spacial domain for the input:
$p = 2, \{xa\} = \{0.5, 0.7, 1.5\}, \{\xi a\} = \{-1, 0, 1\}$